

HCAHPS Patient Analysis Project

Data Quality Assessment Report

Prepared By

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Tools Used

Power BI

Power Query

Excel

Submission Date

6 August 2026

Data Quality Assessment

This report presents the data quality assessment of the HCAHPS Patient Analysis dataset using Power Query in Power BI. The dataset was checked to identify common data quality issues before creating the dashboard. The assessment focused on missing values, duplicate records, data consistency, invalid values, and data type issues.

Issue | Business Impact | Resolution

Issue	Business Impact	Resolution
Missing Values	Missing values can lead to incomplete analysis and inaccurate dashboard results.	No missing (Null) values were found in any of the seven tables. No action was required.
Duplicate Records	Duplicate records can produce incorrect calculations and inaccurate reports.	No duplicate records were found in any of the seven tables. No action was required.
Invalid State Codes	Invalid state codes can affect state-wise analysis and reporting.	All state codes were verified against the States table. No invalid state codes were found.
Missing survey responses	The Completed Surveys and Response Rate (%) columns contain 5,773 records with the value " Not Available ". This may affect survey response analysis.	These values were identified and documented. They should be handled appropriately during data preparation before analysis.
Inconsistent reporting periods	Inconsistent reporting periods can affect trend analysis and reporting.	All Release Period values were verified and found to be consistent across the related tables.
Invalid percentage values	Percentage values outside the valid range can produce incorrect analysis.	All percentage values were verified and found to be between 0 and 100 .
Data type mismatches	The Response Rate (%) column contains numeric values together with " Not Available ", so Power Query imports it as Text . This may affect calculations.	The issue was identified and documented. During data preparation, " Not Available " values can be handled according to business requirements before changing the data type to a numeric format.

Conclusion

The data quality assessment was completed using Power Query. Most of the dataset was clean and consistent. The identified issues were documented for future data preparation and analysis.